

What theoretical guarantees of correctness, completeness, and verifiability provided by automata-based grammar parsers are lost when modern NLP systems rely on probabilistic language models instead of explicit formal grammars?

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Abstract

Automata-based grammar parsers provide formal guarantees of correctness, completeness, and verifiability because they implement explicit formal grammars. This paper asks what happens to those guarantees when modern NLP systems replace automata-based parsers with probabilistic language models. Drawing on work in formal language theory, algebraic topology, neural architecture analysis, and empirical NLP evaluation, we identify three categories of loss: structural, representational, and behavioral. Structurally, under realistic constraints, standard neural architectures fall below the context-free level in the Chomsky hierarchy and therefore cannot support PDA-level guarantees. Representationally, neural networks transform algebraic language structure into continuous geometric approximations that discard the invariants on which decidability and completeness proofs rely. Behaviorally, the absence of formal guarantees leads to unpredictable, error-prone outputs whose failure modes cannot be bounded or characterized from first principles. To ground these claims, we conduct a controlled syntactic stress-test comparing a finite-state automaton (FSA), a context-free grammar (CFG) parser, and a probabilistic language model across six syntactic pattern types at increasing structural depths. The CFG parser maintains formally correct accept/reject behavior across all depths tested, whereas the language model's surprisal signal collapses for many patterns and fails to track grammatical boundaries. Taken together, these findings show that the shift from automata-based to probabilistic NLP is not merely an implementation choice but a fundamental change in what a system can be proven to do.

1: Introduction

The history of natural language processing can be divided into two paradigms. The first, rooted in formal linguistics and automata theory, treats language as a structured object governed by explicit rules. Systems in this paradigm—finite-state transducers, chart parsers, Earley parsers, and CYK implementations of CFGs—derive their behavior directly from grammars, and their correctness can be proven mathematically. The second, now dominant, treats language as a statistical object to be approximated from data: neural language models such as LSTMs, GPT-2, and BERT learn continuous

representations from large text corpora and make predictions based on learned patterns rather than explicit rules.

The transition from the first paradigm to the second has produced dramatic improvements on empirical benchmarks. Empirical performance, however, is not the same as formal correctness. A system can generate fluent, accurate-looking output while lacking any provable relationship to the grammar of the language it models. This gap between approximation and guarantee is the central focus of our paper.

We address our research question in two stages. First, we review five papers spanning formal language theory, algebraic topology, neural architecture analysis, direct parser comparison, and empirical robustness testing. Second, we design a controlled syntactic stress-test that makes the predicted losses observable and measurable. Our goal is not to argue that neural models are practically useless, but to characterize what is sacrificed when formal guarantees are traded for statistical flexibility.

The remainder of the paper is organized as follows. Section 2 reviews relevant literature. Section 3 synthesizes the literature into three categories of loss. Section 4 describes our modeling methodology. Section 5 presents preliminary results. Section 6 discusses their implications, and Section 7 concludes.

2: Literature Review

To understand which guarantees are lost when automata-based parsers are replaced by probabilistic language models, we need to clarify three points: (1) what guarantees automata-based parsers provide, (2) why standard neural architectures cannot preserve those guarantees structurally, and (3) what happens empirically when the guarantees are absent.

2.1 Formal Guarantees of Automata-Based Parsing

Automata-based approaches to NLP rest on a well-defined theoretical foundation. Kamble et al. (2024) survey how finite-state automata, pushdown automata, and Turing machines support tasks such as tokenization, parsing, and syntactic analysis. At the core of this foundation is a set of provable properties: determinism, decidability, soundness, and completeness. These properties are not incidental; they follow necessarily from the explicit grammar rules that govern the computation. Kamble et al. acknowledge scalability challenges for automata-based models on large datasets and discuss trade-offs between formal guarantees and practical flexibility, but they treat the guarantees themselves as an important baseline even in a statistical era.

Oh et al. (2022) make this baseline concrete by directly comparing structural parsers with neural language models. Structural parsers based on probabilistic context-free grammars compute probabilities over explicit parse trees whose nodes are labeled with syntactic categories and whose derivations are governed by grammar rules. Even when a neural language model matches or exceeds a CFG-based parser in surprisal, only the parser is provably sound and complete with respect to its grammar. The neural model

has no such guarantee and can only be assessed output by output. This distinction between formal provability and empirical approximation motivates the rest of our analysis.

2.2 How Neural Architectures Map to Formal Language Classes

Automata-based parsers derive their guarantees from their position in the Chomsky hierarchy, so a natural question is where neural architectures land in the same hierarchy. Ackerman and Cybenko (2020) study CNNs, SRNNs, GRUs, LSTMs, Transformers, and memory-augmented networks under realistic constraints such as linear-time computation and finite-precision states. They show that simple recurrent networks and GRUs are effectively limited to regular languages, the weakest class. LSTMs can simulate simplified counter machines and thus recognize some richer “counter” languages, but still fall short of the full context-free class recognized by PDAs. Transformers with periodic positional encodings are asymptotically weaker than full regular language recognition, and memory-augmented networks can approach context-sensitive power only with substantial training difficulty.

For our purposes, this means that a PDA implementing a CFG remains provably sound and complete for that grammar, whereas an LSTM or Transformer, operating within weaker language classes, cannot make the same claim for genuinely context-free phenomena. The gap between what a neural model can approximate and what an automaton can guarantee is therefore not a matter of training or data scale; it is a structural difference in expressive capacity.

2.3 The Representational Transformation Underlying the Loss

The structural limitations documented by Ackerman and Cybenko (2020) have a deeper mathematical explanation. In related work, Ackerman and Cybenko (2021) analyze how neural networks change formal languages at the level of internal representation using algebraic topology. Formal languages have specific topological signatures; for example, the Dyck-2 language of balanced parentheses over two bracket types has the topology of a figure-eight with exactly two loops, a fixed and provable property of its grammar.

When neural networks are trained on such languages, Ackerman and Cybenko track the evolution of this topology layer by layer using persistent homology. They find that networks systematically convert formal languages from algebraic structures into continuous point clouds that are progressively simplified until linearly separable. The topological invariants that encode the grammar’s structure are not preserved; they are discarded as a side effect of optimization. This means the loss of guarantees is not just about probabilistic versus deterministic computation. Once algebraic structure is replaced by a geometric approximation, there is no remaining object from which to derive proofs of decidability, completeness, or length-independent generalization.

2.4 Empirical Consequences When Guarantees Are Absent

These theoretical losses have practical consequences. Miaschi et al. (2022) study BERT’s behavior on Italian learner essays containing grammatical, orthographic, and lexical errors. Despite massive training data, BERT shows substantial sensitivity to these errors: types of errors that humans easily accommodate cause measurable degradation in BERT’s internal representations, affecting both semantic similarity judgments and syntactic probing tasks. This behavior is consistent with a system lacking formal guarantees. An automaton with an explicit grammar responds deterministically to grammatical and ungrammatical inputs, accepting or rejecting according to clear rules. A neural model has no such fixed behavior, and the pattern of degradation cannot be bounded or predicted in advance.

Oh et al. (2022) add a complementary perspective. Even when a neural language model and a structural parser achieve similar surprisal scores, they do so via different mechanisms: the parser operates over explicit trees and grammar rules, while the neural model relies on learned continuous representations. Matching predictive accuracy does not restore formal guarantees; it does not ensure that all valid strings are handled correctly or that errors on invalid strings are systematic.

3: Analysis

The literature converges on a single answer to our research question. The shift from automata-based parsers to probabilistic language models produces three distinct categories of loss: structural, representational, and behavioral.

3.1 Structural Loss: Capacity Below the Context-Free Level

Automata-based parsers are defined by their position in the Chomsky hierarchy: FSAs recognize regular languages, PDAs recognize context-free languages, and Turing machines recognize recursively enumerable languages, each with associated recognition guarantees. A PDA-based parser for a CFG is guaranteed to accept every valid string and reject every invalid one. Ackerman and Cybenko (2020) show that under realistic constraints, standard neural architectures cannot occupy the same positions in this hierarchy. SRNNs and GRUs are limited to regular languages, LSTMs reach only counter languages, and Transformers fall short of full regular recognition, so none of these architectures can provide PDA-level guarantees for genuinely context-free structures such as nested dependencies, center-embedding, or balanced brackets.

3.2 Representational Loss: From Algebraic Structure to Geometric Approximation

Even when neural models are trained directly on formal languages, the way they represent those languages internally is fundamentally different from automata. Ackerman and Cybenko (2021) show that formal languages have precise topological structures—again, Dyck-2 has the topology of a figure-eight—that are invariant properties of the underlying grammar. Neural networks trained on these languages convert them into continuous point clouds that become linearly separable by the final

layers. Preserving algebraic invariants is incompatible with this drive toward linear separability, so the invariants are destroyed.

This transformation directly affects correctness and verifiability. A formal parser can be verified because its behavior follows from an explicit grammar with mathematically characterizable properties. A neural model’s behavior follows from learned weights whose connection to any formal language is at best approximate. The model does not implement a grammar; it approximates a distribution over strings. Without an explicit formal object to compare against, there is nothing to verify.

3.3 Behavioral Loss: Unpredictability Under Grammatical Uncertainty

If a system lacks the structural capacity to recognize a formal language and its representations no longer encode formal structure, its behavior on grammatical violations cannot be predicted or bounded from first principles. Miaschi et al. (2022) show that BERT’s internal representations degrade in complex, non-systematic ways when faced with learner errors, and there is no simple rule system that predicts which errors will cause which failures. An automaton’s response to ungrammatical input, by contrast, is completely predictable: it rejects the string at the point where the grammar is violated.

Oh et al. (2022) further show that when a neural language model and a structural parser achieve similar surprisal on in-distribution data, their behavior diverges once inputs move away from the training distribution. The neural model’s errors are opaque and distribution-dependent, while the parser’s errors, when they occur, are traceable to gaps in the grammar. The system works until it does not, and there is no principled way to locate that boundary in advance.

4: Methodology

To make these three categories of loss concrete, we construct an alpha-version syntactic stress-test. On the formal side, we implement a hand-built finite-state automaton for shallow Dyck-1 and simple subject–verb agreement, a pushdown automaton for Dyck-1 and Dyck-2, and a CFG with an Earley parser that covers singular/plural agreement, relative clauses, and simple center-embedding. On the probabilistic side, we use a language model evaluated via surprisal, standing in for a modern neural LM. We generate controlled sets of grammatical and minimally ungrammatical strings for six pattern types, spanning regular, counter-like, and context-free structures, and increasing nesting depth. For each string we record binary accept/reject decisions from the automata and CFG parser and token-level surprisal from the language model.

5: Results

In Dyck-1, the PDA is correct on all 8 test strings, while the FSA succeeds only on 5, failing exactly when nesting depth exceeds one unmatched opening symbol; the depth-scaling experiment confirms this, with PDA accuracy at 12/12 versus 7/12 for the FSA as depth increases. In Dyck-2, the PDA achieves 6/6 accuracy and sharply separates well-formed from ill-formed bracket strings, whereas the language model assigns similar surprisal to valid and invalid sequences and never rejects any input. In subject–verb agreement and center-embedding, the CFG parser correctly classifies 8/10 sentences, with its two errors confined to long-distance agreement cases that can be repaired by extending the grammar, while the language model shows only a small surprisal gap between grammatical and ungrammatical variants and again offers no hard accept/reject boundary. Summarizing across experiments, the FSA and CFG/PDA models preserve soundness, completeness (within their grammar), and verifiability, whereas the neural language model lacks a reject state, has no completeness guarantee, and provides only opaque, statistical signals about grammaticality.

6: Discussion

These alpha experiments make the three categories of loss from our literature review visible in a concrete setting. Structurally, the Dyck-1 and depth-scaling results show that finite-state machinery fails exactly where stack-like memory is required, while the PDA remains perfectly accurate across all tested depths; this mirrors Ackerman and Cybenko’s claim that context-free guarantees require PDA-level power and cannot be matched by weaker architectures. Representationally, the CFG and PDA make discrete accept/reject decisions grounded in explicit rules, so every error can be traced to a missing or mis-specified production, whereas the language model’s surprisal values blur the line between grammatical and ungrammatical strings and provide no principled notion of “the” language being recognized. Behaviorally, the contrast between deterministic parser outputs and the language model’s smooth but shallow surprisal gaps illustrates the verifiability gap documented by Oh et al. and Miaschi et al.: formal parsers offer predictable, auditable failure modes, while neural models degrade in pattern-specific, distribution-dependent ways that cannot be bounded in advance.

At the same time, the experiments also expose limits on the formal side. Our CFG misses two long-distance agreement cases, underscoring that completeness is conditional on the coverage of the grammar rather than guaranteed “for English” as a whole. However, those failures are transparent: we can see exactly which constructions are missing and, in principle, repair them by extending the rule set. No analogous repair path exists for the language model, whose behavior on the same inputs is determined by high-dimensional weight interactions rather than explicit rules. This asymmetry reinforces our central claim: empirical competence does not by itself recover the guarantees that follow from explicit grammar implementation.

7: Conclusion

Our goal was to determine which guarantees of correctness, completeness, and verifiability are lost when automata-based grammar parsers are replaced by

probabilistic language models. The literature shows that standard neural architectures occupy weaker positions in the Chomsky hierarchy, collapse algebraic language structure into geometric approximations, and exhibit opaque, distribution-sensitive failure modes. Our alpha stress-test connects these theoretical claims to observed behavior: PDAs and CFGs maintain sharp grammatical boundaries and auditable error profiles, while the language model assigns probabilities to every string without a principled notion of rejection or coverage.

These findings support a nuanced view of modern NLP. Probabilistic language models offer impressive empirical performance, but they do so by abandoning the structural, representational, and behavioral conditions that make formal guarantees possible. Automata-based components still provide unique value wherever correctness and verifiability matter, even if they must be combined with neural modules to reach practical scale. Future work will replace our toy language model with large pretrained LMs, extend the grammar to richer forms of center-embedding and tense, and explore hybrid architectures that explicitly separate a provable formal core from a probabilistic outer layer, rather than treating neural models as drop-in replacements for parsers with known guarantees.

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